

ISE 5405: Optimization I

Lecture 5: The Geometry of Linear Programming II

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Bases, Degeneracy, Existence & Optimality • Bertsimas–Tsitsiklis
§§2.3–2.8

- **Corners in standard form:** bases, basic variables, and how to compute a BFS §2.3
- **Degeneracy:** when too many constraints pile onto one point §2.4
- **Existence:** which polyhedra have corners at all §2.5
- **Optimality:** the theorem that lets us search only corners §2.6
- **Two representations + Fourier–Motzkin:** polytopes as convex hulls; eliminating variables §§2.7–2.8

Destination

A finite candidate set, an adjacency structure to walk on, and a guarantee that the walk is worth taking: the simplex method will be fully motivated.

Polyhedra in Standard Form (§2.3)

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}, \quad A \in \mathbb{R}^{m \times n}.$$

Full-row-rank setup

The m rows of A are linearly independent (full row rank), so $m \leq n$. If not, Theorem 2.5 says that, whenever $P \neq \emptyset$, deleting dependent rows leaves the same polyhedron.

Counting intuition

$Ax = b$ uses m degrees of freedom, leaving an $(n - m)$ -dimensional world. A corner needs those remaining freedoms pinned by $n - m$ constraints $x_i = 0$. Thus a corner amounts to choosing which variables may remain nonzero.

Basic Solutions in Standard Form (Thm 2.4)

Theorem

x is a basic solution of $\{x : Ax = b, x \geq 0\}$ if and only if $Ax = b$ and there are indices $B(1), \dots, B(m)$ such that:

1. columns $A_{B(1)}, \dots, A_{B(m)}$ are linearly independent; and
2. $x_i = 0$ for every $i \notin \{B(1), \dots, B(m)\}$.

The recipe

1. Choose m independent columns of A : the **basis**.
2. Set every other, **nonbasic**, variable to zero.
3. Solve $Bx_B = b$, equivalently $x_B = B^{-1}b$, for the **basic** variables.

Feasibility decides

If $x_B \geq 0$, this is a BFS. A negative entry makes it basic but infeasible: a corner of the algebra, not of P . Here $B = [A_{B(1)} \cdots A_{B(m)}]$ is the basis matrix and $B(i)$ are the basic indices.

Pick a Basis, Get a Corner (or Not)

$$\begin{bmatrix} 1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 6 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 8 \\ 12 \\ 4 \\ 6 \end{bmatrix}, \quad x \geq 0.$$

$n = 7$, $m = 4$: choose four columns. The static table reveals the basis chooser's outcomes.

Basis	solution x	verdict
Initial state: select exactly four columns.		
$\{4, 5, 6, 7\}$	$(0, 0, 0, 8, 12, 4, 6)$	BFS, nondegenerate
$\{3, 5, 6, 7\}$	$(0, 0, 4, 0, -12, 4, 6)$	basic, infeasible
$\{1, 2, 3, 4\}$	$(4, 6, 1, -4, 0, 0, 0)$	basic, infeasible

The simplex data structure

Different bases give different corners; singular choices give no basic solution; negative components give basic but infeasible solutions.

Degeneracy (§2.4)

Definitions 2.10 and 2.11

A basic solution is **degenerate** if more than n constraints are active there.

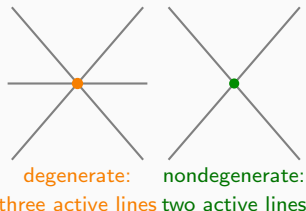
In standard form, this means more than $n - m$ components are zero: at least one basic variable is also zero.

BT example ($n = 3$).

- $x = (2, 6, 0)$: exactly three active constraints; nondegenerate.
- $x = (4, 0, 2)$: four active constraints; degenerate.

Why care?

Different bases can describe the same degenerate corner. An algorithm may stall while changing bases without moving—the seed of cycling in Chapter 3.



Degeneracy is rare in random data but common in structured models.

When Do Corners Exist? (§2.5)

Not every polyhedron has a corner: a halfplane in $\mathbb{R}^2$ has none.

Definition 2.12. P contains a line if some $\bar{x} \in P$ and $d \neq 0$ satisfy

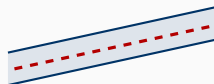
$$\bar{x} + \lambda d \in P \quad \text{for every } \lambda \in \mathbb{R}.$$

Theorem 2.6. For nonempty $P = \{x : a_i^\top x \geq b_i\}$, these are equivalent:

1. P has an extreme point;
2. P contains no line;
3. n of the vectors a_i are linearly independent.

Corollary 2.2

Nonempty bounded polyhedra and nonempty standard-form polyhedra always have a BFS. In standard form, $x_i \geq 0$ supplies n independent vectors.



contains a line; no corners



no line; corners exist

Extreme Point $\Rightarrow$ No Line

Claim

If P has an extreme point x^{ext} , then P contains no line.

1. Suppose instead that $\bar{x} + \lambda d \in P$ for all $\lambda \in \mathbb{R}$, where $d \neq 0$.
2. For $P = \{x : Ax \geq b\}$, a full line implies $Ad = 0$: otherwise one sign of sufficiently large λ violates some row. Therefore, from every $x \in P$, both $x + d$ and $x - d$ remain feasible.
3. In particular $x^{\text{ext}} + d, x^{\text{ext}} - d \in P$, and

$$x^{\text{ext}} = \frac{1}{2}(x^{\text{ext}} + d) + \frac{1}{2}(x^{\text{ext}} - d).$$

The two endpoints are distinct from x^{ext} .

Contradiction

The displayed midpoint representation violates extremality. Thus P contains no line.

□ The reverse directions use the rank condition.

Optimality of Extreme Points (§2.6)

Theorem 2.7

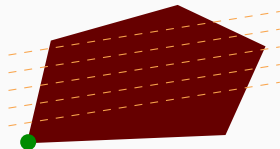
If P has an extreme point and $\min\{c^T x : x \in P\}$ has an optimal solution, then **some optimal solution is an extreme point**.

Theorem 2.8 (stronger)

If P has an extreme point, then either the optimal cost is $-\infty$, or an extreme point is optimal.

Corollary 2.3 $\Rightarrow$ Property 1

Every LP over a nonempty polyhedron either has cost $-\infty$ or attains an optimum. No e^x -style pathology: LP keeps the trichotomy exact.



Last contact: a corner or an edge containing corners.

Some Optimal Solution is an Extreme Point

Let $v = \min_{x \in P} c^\top x$, attained by assumption, and let

$$Q = \{x \in P : c^\top x = v\}$$

collect every optimum. The strategy is to find an extreme point of Q and promote it to one of P .

1. Q is a polyhedron: add $c^\top x \leq v$ and $c^\top x \geq v$ to the rows defining P .
2. P has an extreme point, hence contains no line. The nonempty subset Q contains no line, so Theorem 2.6 gives an extreme point x^* of Q .
3. Suppose $x^* = \lambda y + (1 - \lambda)z$, where $y, z \in P \setminus \{x^*\}$ and $\lambda \in (0, 1)$.
4. Since $c^\top y, c^\top z \geq v$ and

$$\lambda c^\top y + (1 - \lambda) c^\top z = c^\top x^* = v,$$

positive weights force $c^\top y = c^\top z = v$. Thus $y, z \in Q$, contradicting extremality of x^* in Q .

Conclusion. No witnesses in P exist: x^* is extreme in P , and because $x^* \in Q$, it is optimal. $\square$

The Other Representation: Convex Hulls (§2.7)

Definition 2.5

A **convex combination** of $x_1, \dots, x_k$ is

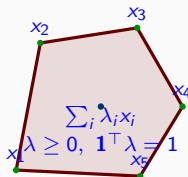
$$\sum_{i=1}^k \lambda_i x_i, \quad \lambda_i \geq 0, \quad \sum_{i=1}^k \lambda_i = 1.$$

The **convex hull** is the set of all such combinations.

Theorem 2.9

A nonempty polytope is the convex hull of its extreme points: outside-in (inequalities) and inside-out (corners) describe one object.

Inequalities support feasibility checks; corner combinations support proofs and decompositions.



Static hull-builder status

The five boundary points are extreme. The blue interior point is already in their hull, so it is not needed to generate it.

Fourier–Motzkin: Solving by Shadows (§2.8)

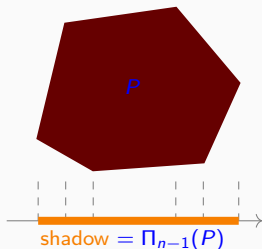
Question: is $P = \{x : Ax \geq b\}$ empty?

Projection

$\Pi_{n-1}(P)$ is the shadow after forgetting x_n .

Crucially,

$$P \neq \emptyset \iff \Pi_{n-1}(P) \neq \emptyset.$$



- Eliminate one variable at a time:
 $\mathbb{R}^n \rightarrow \mathbb{R}^{n-1} \rightarrow \dots \rightarrow \mathbb{R}.$
- In $\mathbb{R}$, check whether an interval $[l, u]$ with $l \leq u$ remains.
- **Theorem 2.10:** the eliminated system is exactly the projection. Hence projections of polyhedra are polyhedra, and (Cor. 2.6) convex hulls of finite point sets are polyhedra.

The Elimination Step

To eliminate x_n , sort constraints by the sign of its coefficient, normalize, and rewrite:

$$\begin{aligned}x_n &\geq d_i + f_i^\top \bar{x} & i \in I^+ & \text{(lower bounds),} \\d_j + f_j^\top \bar{x} &\geq x_n & j \in I^- & \text{(upper bounds),} \\0 &\geq d_k + f_k^\top \bar{x} & k \in I^0 & \text{(x_n absent).}\end{aligned}$$

An x_n exists iff every upper bound dominates every lower bound. Thus add

$$d_j + f_j^\top \bar{x} \geq d_i + f_i^\top \bar{x} \quad (i \in I^+, j \in I^-)$$

and keep the I^0 rows. The variable x_n is gone.

Cost

The step removes $|I^+| + |I^-|$ rows and adds $|I^+||I^-|$. Constraint counts can grow exponentially: beautiful theory, poor algorithm.

Why teach it?

Every derived inequality is a nonnegative combination of original rows—the seed of Farkas' lemma and LP duality in Chapter 4.

Fourier–Motzkin in Action

Start with seven inequalities in three variables:

$$\begin{array}{lll} (1) \quad -x_1 \geq -1 & (2) \quad -x_2 \geq -1 & (3) \quad -x_3 \geq -1 \\ (4) \quad -x_1 - x_2 \geq -3 & (5) \quad -x_1 - x_3 \geq -3 & (6) \quad -x_2 - x_3 \geq -3 \\ (7) \quad x_1 + x_2 + x_3 \geq 6. \end{array}$$

Eliminate x_3 . Row 7 is the lower bound; rows 3, 5, 6 are upper bounds:

$$x_3 \geq 6 - x_1 - x_2, \quad x_3 \leq 1, \quad x_3 \leq 3 - x_1, \quad x_3 \leq 3 - x_2.$$

Pairing lower with upper yields

$$x_1 + x_2 \geq 5, \quad x_2 \geq 3, \quad x_1 \geq 3.$$

But carried row 1 says $x_1 \leq 1$. The shadow is empty, so

$$P = \emptyset. \quad \square$$

Why is LP Special? The Three Properties, Earned

Property 1

Every LP is infeasible, unbounded, or has an optimal solution—no in-between.

Cor. 2.3

Property 2

Every local optimum is global: linear objectives over convex sets cannot trap you.

Property 3

A nonempty polytope has an optimal extreme point.

Thm 2.7/2.8

The simplex method's birth certificate

Finitely many candidate corners + an edge structure to walk + a guarantee that the best corner is globally best. Chapter 3 asks how to walk cleverly.

Key Takeaways

- Standard-form corners are **bases**: choose m independent columns, zero the rest, solve, and test feasibility.
- **Degeneracy**: extra active constraints; many bases, one point; algorithmic headaches ahead.
- **Existence** (Thm 2.6): corners $\iff$ no line inside $\iff n$ independent constraint vectors.
- **Optimality** (Thm 2.7/2.8): searching corners is enough, and Property 1 follows.
- **Representations**: inequalities or convex hull of corners; Fourier–Motzkin projects and foreshadows duality.

Next: the Simplex Method

We finally get to walk the corners.